EXHIBITION PROJECT

# Obstacle Avoidance Car

*A car that sees before it moves*



PHOTO: finished car


**Credits**

Tijil Das · Shreyam Chaudhury · Shreyas Chatterjee · Shaurya Gupta

St. Aloysius School -· Class 12 Computer Science Exhibition

DID YOU KNOW

# Nature invented it first



**BAT**

Echo = eyes in the dark



**DOLPHIN**

Clicks bounce off prey

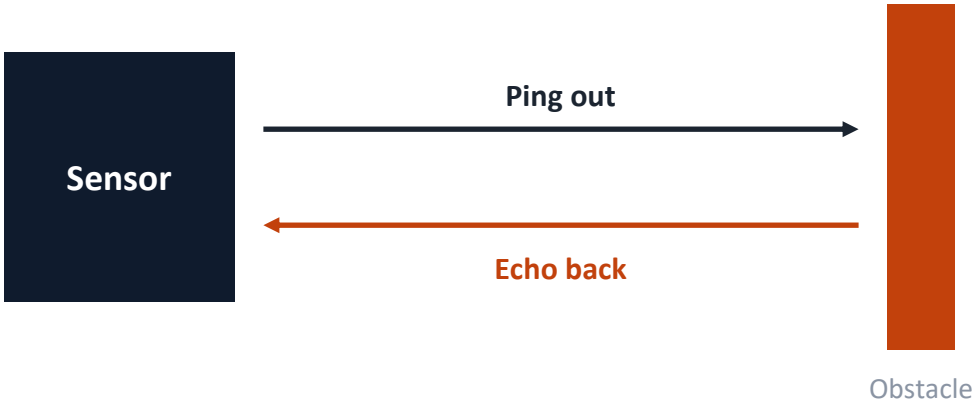


**OUR CAR**

40 kHz pings bounce off obstacles

*Three bodies. One idea: listen to the echo.*

# How it sees




**Distance = (Time × Speed of sound) ÷ 2**

**40 kHz**

A silent shout. Humans hear only up to about 20 kHz.

**~0.6 ms**

Echo time from an obstacle 10 cm away.

HOW IT WORKS

# What's inside

**Ultrasonic Sensor + Servo**

1 to 2 ms pulses set its angle

→

**Arduino Uno R3**

The brain

→

**L298N Motor Driver**

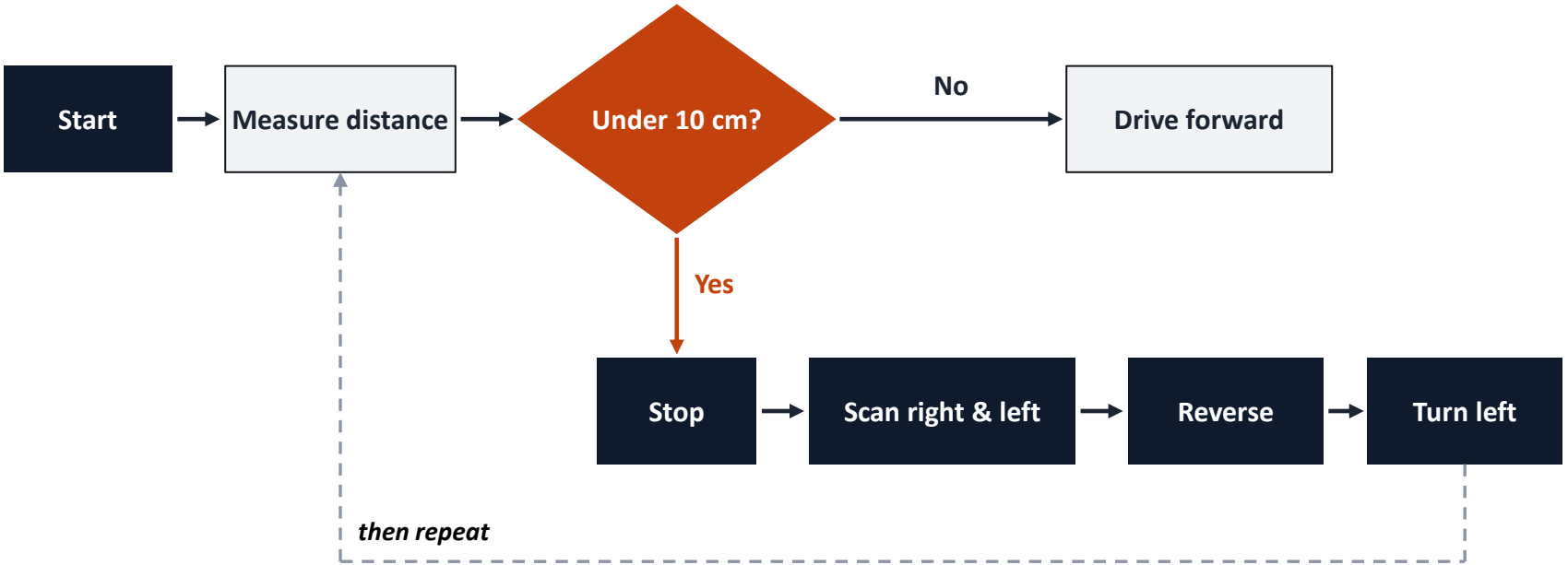
Flips current to reverse

→

**2 Geared DC Motors**

**9V Battery** powers everything, with a shared ground

# How it decides

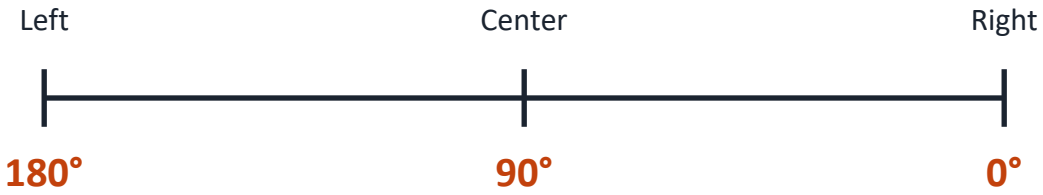


HOW IT WORKS

# When it meets an obstacle

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| **Stop** | **Scan right** | **Scan left** | **Center** | **Reverse** | **Turn left** |

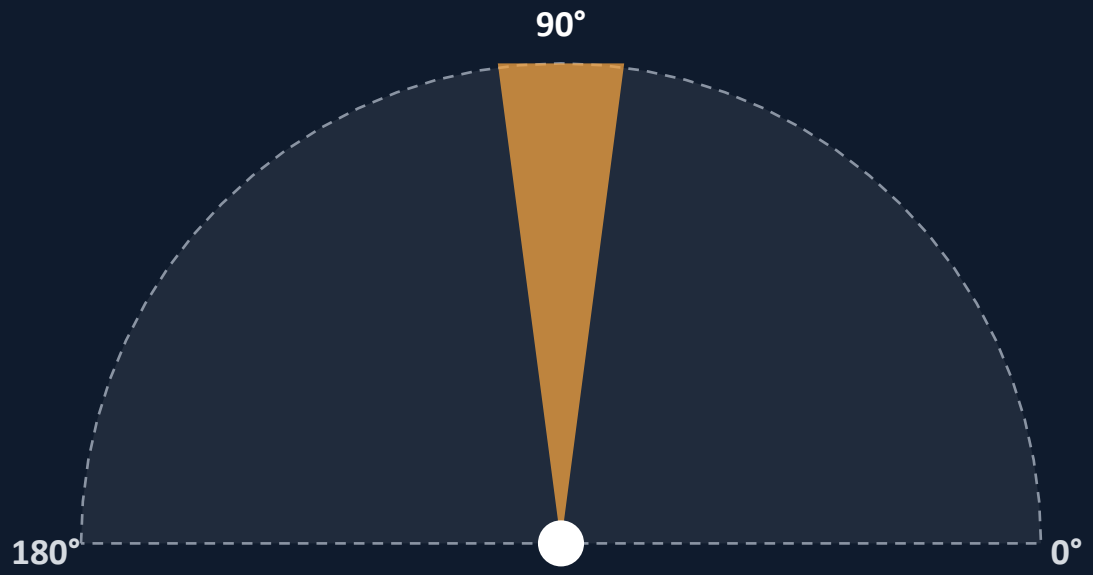
SERVO ANGLE




*Then it drives on.*

DID YOU KNOW

# A flashlight, not a camera




The sensor sees only a narrow cone (about 15°).

**So a servo sweeps it from 0° to 180°.**

DID YOU KNOW

# Can you fool it?



SOFT CLOTH

**Absorbs the ping**



STEEP ANGLE

**Bounces the echo away**



THIN POLE

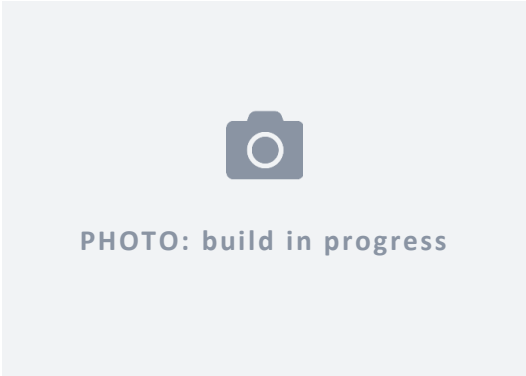
**Barely echoes**

**Even a hot room changes the speed of sound.**

That's why real robots combine several sensors.

THE BUILD

# Building it




[Caption]




[Caption]




[Caption]

**CHALLENGE** **Arduino reset whenever motors started moving**

→

**FIX** Separated motor power from Arduino logic.

THE RESULT

# By the numbers

**10 cm**
Obstacle trigger

**π rads**
Servo sweep

**40 kHz**
Silent ping

**2**
Drive motors

# Thank you

Ask us anything.

# QUICK QUIZ

QUICK QUIZ Q1 of 5

# Our distance formula divides the time by 2. Why?

A The echo slows down on its way back

B The sound travels to the obstacle AND back

C The car has two sensors

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A The echo slows down on its way back

B **The sound travels to the obstacle AND back** ✓

C The car has two sensors

The timer counts the whole round trip, but we want one-way distance.

QUICK QUIZ Q2 of 5

# Two identical obstacle-avoiding cars run side by side at this exhibition. What could go wrong?

A They may hear each other's pings and see a "phantom obstacle"

B Their batteries drain twice as fast

C Their sound waves cancel and both go silent

QUICK QUIZ Q2 of 5

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Echo sensors can't tell whose ping is whose, so they get confused.

QUICK QUIZ Q3 of 5

# An echo returns after 10 milliseconds. Sound travels at about 340 m/s. How far is the obstacle?

QUICK QUIZ Q3 of 5

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✓ **About 1.7 metres**

340 × 0.01 = 3.4 m there and back, so divide by 2.

Hint was in Q1: only half the trip counts.

QUICK QUIZ Q4 of 5

# Our car treats under 10 cm as an obstacle but needs over 15 cm to call the path clear. Why leave a gap?

A So readings that jitter by a centimetre or two don't make it flip-flop

B Because the sensor can't measure between 10 and 15 cm

C To save battery

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A gap, called a buffer zone, keeps decisions stable.



QUICK QUIZ Q5 of 5

# Why does the sensor measure distance with sound, not light?

A Sound is cheaper to make

B Sound is slow enough for a simple Arduino to time the echo

C Light can't bounce off obstacles

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At 10 cm, a light echo takes under a billionth of a second.